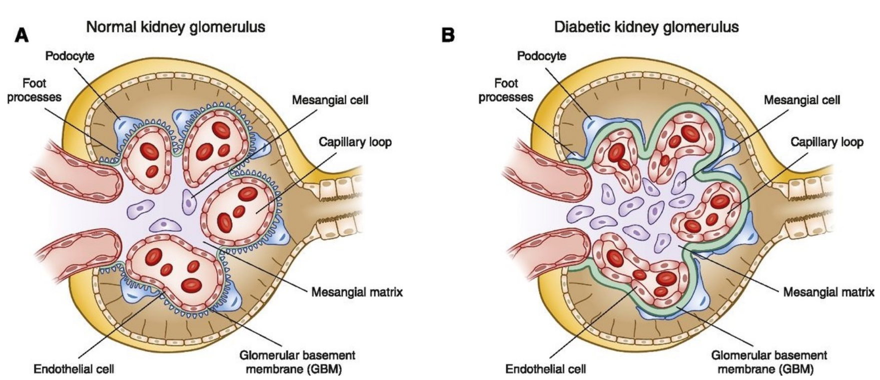
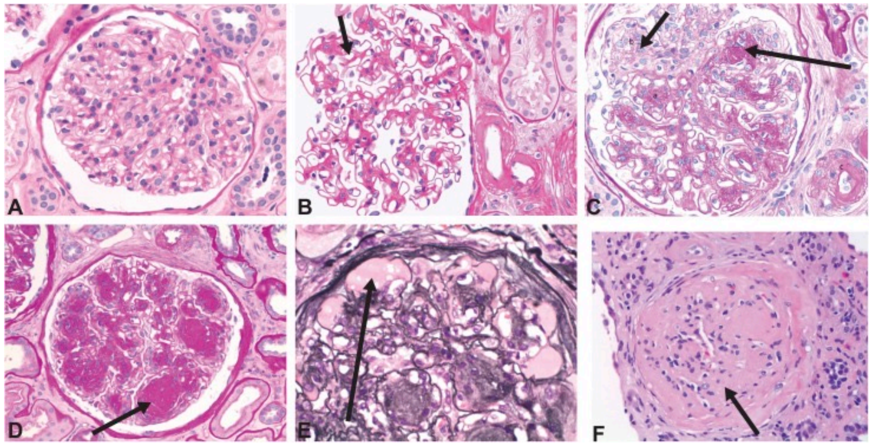
2026/FP03 – Glomeruli detection

- **Project Owner**: Francesco Ponzio - *francesco.ponzio@polito.it*
- **Project Outline**: A correct identification and classification of glomeruli if of paramount importance for the prognosis of the diabetic kidney disease [1]. This project aims at developing a ML pipeline able to: (i) identify the glomeruli; (ii) segment them; (iii) separate different glomeruli classes in an unsupervised way. Since no ground truth is available in terms of glomeruli classification, we wish an unsupervised solution to see if any separability exists. Consider manifold learning methods to assess the separability of the different level of glomeruli necrotization (see below).





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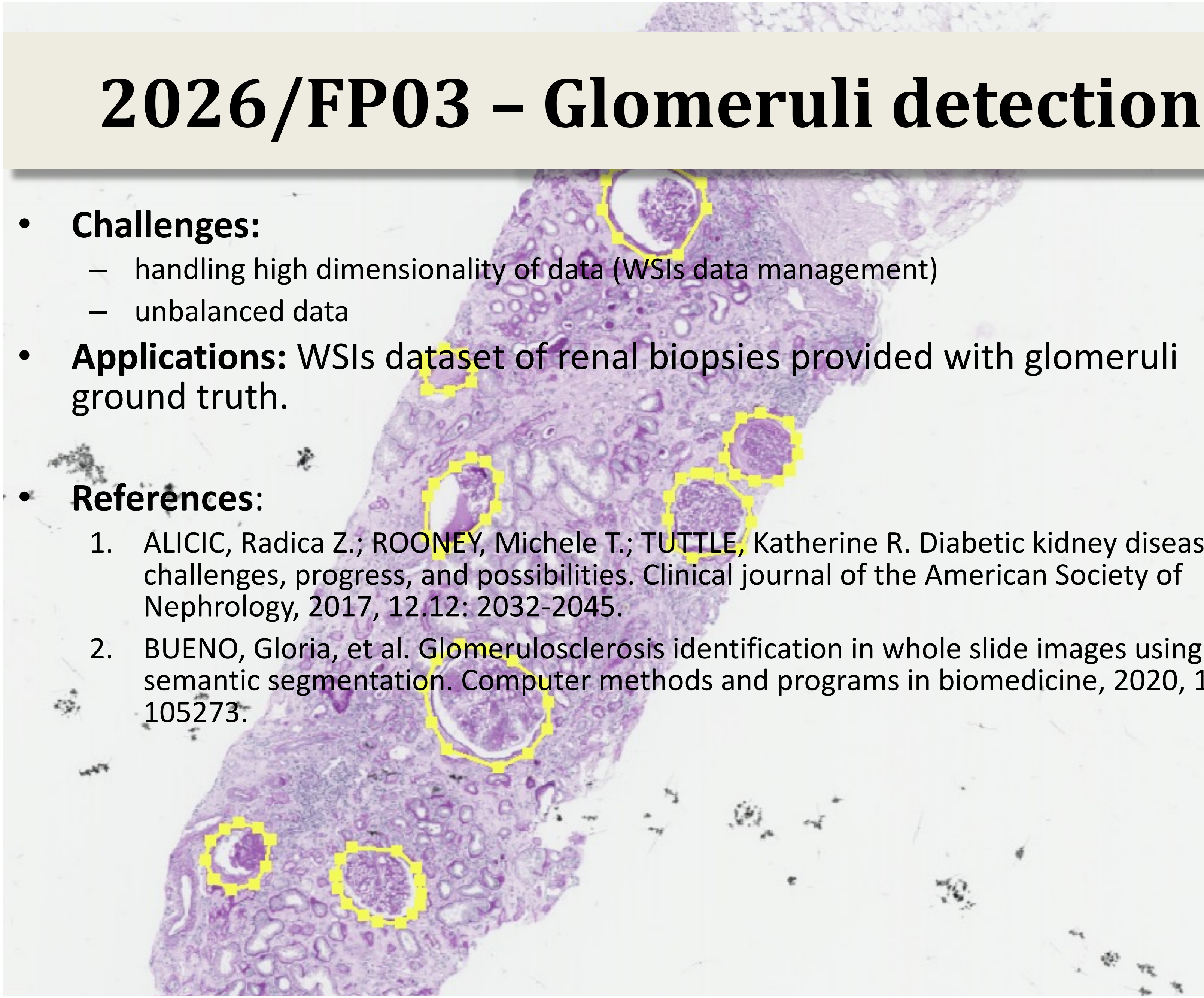

- **Challenges:**
  - handling high dimensionality of data (WSIs data management)
  - unbalanced data
- **Applications:** WSIs dataset of renal biopsies provided with glomeruli ground truth.

- **References**:

1. ALICIC, Radica Z.; ROONEY, Michele T.; TUTTLE, Katherine R. Diabetic kidney disease: challenges, progress, and possibilities. Clinical journal of the American Society of Nephrology, 2017, 12.12: 2032-2045.
2. BUENO, Gloria, et al. Glomerulosclerosis identification in whole slide images using semantic segmentation. Computer methods and programs in biomedicine, 2020, 184: 105273.